

- 3 (a) (gravitational potential energy is) the energy/ability to do work of a mass that it has or is stored due to its position/height in a gravitational field B1
- kinetic energy is energy/ability to do work a object/body/mass has due to its speed/velocity/motion/movement B1 [2]
- (b) (i) $s = [(u + v)t]/2$ or acceleration = $9.8/9.75$ (using gradient) C1
- $= [(7.8 + 3.9) \times 0.4]/2$ or $s = 3.9 \times 0.4 + \frac{1}{2} \times 9.75 \times (0.4)^2$ C1
- $s = 2.3(4) \text{ m}$ A1 [3]
- (ii) $a = (v - u)/t$ or gradient of line C1
- $= (7.8 - 3.9)/0.4 = 9.8 (9.75) \text{ m s}^{-2}$ (allow $\pm \frac{1}{2}$ small square in readings) A1 [2]

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(iii) $KE = \frac{1}{2} mv^2$ C1

change in kinetic energy = $\frac{1}{2} mv^2 - \frac{1}{2} mu^2$
 $= \frac{1}{2} \times 1.5 \times (7.8^2 - 3.9^2)$ C1

$= 34 \text{ (34.22) J}$ A1 [3]

(c) work done = force $\times$ distance (moved) or Fd or Fx or mgh or mgd or mgx M1

$= 1.5 \times 9.8 \times 2.3 = 34 \text{ (33.8) J (equals the change in KE)}$ A1 [2]